

Muhammad Hamza Anwaar

Backend Engineer · AI & Voice Automation · Python · FastAPI · Django REST · Open to Remote

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PROFESSIONAL SUMMARY

Backend engineer with 14+ months of production experience building REST APIs, AI automation pipelines, and enterprise voice systems. Shipped 12+ production endpoints and a real-time Workday HR voice bot (Retell AI + GPT-4o + SOAP/XML) used in live environments. Proficient in Python, FastAPI, Django REST Framework, PostgreSQL, Docker, and CI/CD. Experienced with agile workflows, unit testing (pytest), and end-to-end feature ownership. Open to remote and international roles.

WORK EXPERIENCE

Artificizen

Jan 2025 – Present

Backend Developer & AI Automation Engineer · Lahore, Pakistan

- ▶ **Architected** scalable REST APIs using FastAPI and Django REST Framework — 12+ production endpoints with Pydantic validation, JWT authentication, role-based access control, rate limiting, and Swagger/OpenAPI documentation; zero unplanned downtime in production.
- ▶ **Engineered** a production voice automation system (Retell AI + GPT-4o): employees call, speak naturally, and Workday HR records update live via SOAP/XML — fully autonomous, no HR agent required. Includes caller verification, field-level validation, audit logging, and rollback on failed writes.
- ▶ **Designed and deployed** 3 AI automation pipelines integrating OpenAI GPT-4o for task classification, intent routing, and structured data extraction — eliminating manual processing across multiple client workflows.
- ▶ **Containerized** all microservices using Docker and docker-compose; applied agile development practices and CI/CD workflows via Git-based peer review and iterative delivery cycles.
- ▶ **Designed** PostgreSQL schemas with relational modeling, indexing, and ORM query optimization; maintained unit testing and integration testing coverage using pytest across all production endpoints.

Azgard 9 Ltd

Jun 2024 – Aug 2024

Merchant Intern – Marketing Department · Lahore, Pakistan

- ▶ Conducted competitive market analysis for denim product lines across domestic and international export markets.
- ▶ Coordinated with marketing and export teams on customer communication, data management, and cross-border shipment logistics.

PROJECTS · LIVE DEMOS AT HAMZAAANWAAR4.GITHUB.IO

Workday HR Voice Automation System · Retell AI · GPT-4o · SOAP/XML · FastAPI · PostgreSQL · Docker

- ▶ End-to-end autonomous call system: caller speaks → Retell AI captures intent → GPT-4o extracts field values → SOAP/XML payload written live to Workday — no human in the loop.
- ▶ Caller identity verification, field-level data validation, full audit trail, and automatic rollback on failed Workday API calls.

AI Workflow Automation Platform · FastAPI · GPT-4o · PostgreSQL · Pydantic · Docker · Swagger

- ▶ Async FastAPI backend connecting GPT-4o to business logic — task classification, response generation, structured data routing with audit logging; JWT auth, pytest unit testing and integration test suite.

RESTful Task Management API · Django · DRF · JWT · PostgreSQL · pytest · Swagger

- ▶ Full-featured task API: JWT auth, CRUD, filtering, pagination, role-based permissions, soft-delete, versioned endpoints, Swagger docs, and pytest integration tests.

TECHNICAL SKILLS

Backend	FastAPI, Django, Django REST Framework, REST API design, async Python, microservices
Languages	Python (primary), SQL, XML, JavaScript (basic)
AI & Voice	OpenAI GPT-4o, Retell AI, LLM automation, prompt engineering, voice agent pipelines, AI workflow automation
Cloud & DevOps	Docker, docker-compose, Git & GitHub, Linux/Ubuntu CLI, CI/CD pipelines, agile development
Integrations	Workday SOAP/XML Web Services, REST API integration, webhook design, third-party API integration
Databases	PostgreSQL, SQLite, Django ORM, Pydantic, schema design, query optimization, indexing
Testing & Docs	pytest, unit testing, integration testing, Swagger / OpenAPI, Postman
Methodologies	Agile, Git-based code review, API-first design, test-driven development (TDD basics)

EDUCATION

Bahria University

Sep 2021 – Jul 2025

Bachelor of Business Administration (BBA) · Lahore, Pakistan

- ▶ Relevant coursework: Marketing Strategy, Financial Analysis, Business Statistics, Operations Management.
- ▶ Award: Best Business Plan Presentation — strategic innovation in competitive academic evaluation.
- ▶ Languages: English (Professional working proficiency) · Urdu (Native)

CERTIFICATIONS & AWARDS

- ▶ Meta Back-End Developer Certificate — Coursera (in progress)
- ▶ Award: Best Performer of the Quarter – Artificizen
- ▶ FranklinCovey – The 7 Habits of Highly Effective People
- ▶ FranklinCovey – The 4 Essential Roles of Leadership
- ▶ Award: Best Business Plan Presentation – Bahria University